

Assignment 1: Outlier Detection

1 Introduction

In this assignment, you will explore outlier detection on a real multidimensional dataset. The objective is not simply to run an algorithm, but to investigate the detected outliers and try to understand why particular observations are considered unusual. You can deliver the results either as:

- (a) a Jupyter notebook combining code, text, tables, and visualizations in a clear and readable sequence; or
- (b) the source code and report separately, with the report submitted as a PDF containing the explanations, results, and visualizations.

Choose **one dataset** from the list provided for the assignment. Before applying outlier detection, briefly explore the dataset. Describe what the observations represent, the available features, the number of observations and features, whether there are missing values, and which features you select for the analysis. Describe any preprocessing that you perform. Pay particular attention to feature scaling. Explain whether the selected outlier detection method is sensitive to differences in feature scales and how you handle this in your analysis.

2 Local Outlier Factor – Grade G

Apply **Local Outlier Factor (LOF)** to your selected dataset using an existing implementation. You are **not required to implement LOF from scratch**. Briefly explain the main idea behind LOF and what the LOF score tells you about an observation.

2.0.1 Influence of the Neighbourhood Size

An important parameter of LOF is the number of neighbours k . Run LOF using at least **three different values of k** . Compare the results and investigate the following questions:

- Are the same observations identified as strong outliers for different values of k ?
- Which observations change the most when k changes?
- Does a smaller or larger neighbourhood appear more appropriate for your dataset?
- Why?

Visualize the results in an appropriate way. Depending on the dataset, this could, for example, be a two-dimensional projection of the data where the strongest outliers are highlighted, or another visualization that helps you understand the results.

2.0.2 Investigating the Detected Outliers

Select several observations with the highest outlier scores and inspect their original feature values. Discuss why these observations may have received high LOF scores. Which features make them different from other observations? Do they appear to be data errors, rare but valid observations, or potentially interesting cases? Also consider whether there are observations that you expected to be outliers but that LOF did not identify. Your discussion should connect the results of the algorithm to the actual dataset rather than only reporting LOF scores.

3 Comparison of Outlier Detection Techniques – Grade VG

To obtain grade **VG**, complete the requirements for Grade G and extend your investigation by comparing different approaches to outlier detection. Apply the following three techniques:

1. **Local Outlier Factor (LOF)**
2. **Isolation Forest**
3. **one additional outlier detection technique of your choice**

The third technique can, for example, be One-Class SVM, a k -nearest-neighbour distance method, HBOS, or another established outlier detection algorithm. You may use existing implementations of all algorithms.

3.0.1 Comparing the Methods

Apply all three methods to the **same dataset**. Briefly explain how each method determines whether an observation is unusual. Investigate the observations receiving the strongest outlier scores from each method. Compare the results and discuss:

- Which observations are identified as outliers by several methods?
- Which observations are detected by only one method?
- How similar are the results produced by the different algorithms?
- Can you explain why the methods agree or disagree for particular observations?

Create suitable visualizations that make the similarities and differences between the methods visible.

3.0.2 Influence of Parameters

Investigate important parameters for at least **two of the algorithms**. Show how changing these parameters influences the detected outliers. Do not test parameters only to find the combination that produces the “best” result. Instead, discuss why the behaviour of the algorithm changes when its parameters change.

3.0.3 Evaluation and Interpretation

If your dataset contains labels that can reasonably be used as ground-truth information about unusual observations, use suitable quantitative measures to support your evaluation. If suitable ground-truth outlier labels are not available, compare the methods using other evidence, such as overlap between their highest-ranked outliers, differences in outlier rankings, visual analysis, inspection of the original observations, and stability when algorithm parameters change. Finally, answer the following question:

Which outlier detection technique would you choose for this dataset, and why?

Support your conclusion using both the characteristics of the dataset and the observed behaviour of the algorithms.

4 Suggested Datasets

Choose **one** of the following datasets from the UCI Machine Learning Repository:

- Dry Bean – Biology / image-derived features
- MAGIC Gamma Telescope – Physics
- AI4I 2020 Predictive Maintenance – Industrial / predictive maintenance
- Air Quality – Environmental / sensor data
- Cardiotocography – Healthcare

The datasets have been selected because they contain several numerical features and enough observations to make outlier detection meaningful. You do not need to use every available feature. However, if you exclude features from the analysis, briefly explain why.